

Mohammad Raditia Pradana

Researcher, SPARC - Department of Geography,
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M.Sc. in Geography Sciences

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RESEARCH INTERESTS

Street-view imagery (SVI) • Urban data science • Computer vision • Geospatial analysis • Urban green spaces • Urban planning

EDUCATION

M.Sc. in Geography Sciences (Fast-Track Program) — *Universitas Indonesia* 2024 – 2025

GPA **4.00 / 4.00** | Summa Cum Laude | Ranked 1st in Cohort - Faculty level

Thesis: Spatial Modeling of Visual Element Proportion Diversity in Urban Spaces Based on Street-View and Spatial Indicators (Kebayoran Baru & Tanah Abang)

B.Sc. in Geography (Fast-Track Program) — *Universitas Indonesia* 2020 – 2024

GPA **3.85 / 4.00** | Cum Laude | Ranked 1st in Cohort - Department level

Thesis: Protection Level for Fire-Prone Areas in East Jakarta City

PUBLICATIONS

Journal Articles

- [1] **Pradana, M.R.** & Semedi, J.M. (2026). Indonesia's urban green spaces dilemma: Balancing law, ecology, and public space, highlighting Jakarta. *Human Geography*. [\[DOI\]](#)
- [2] **Pradana, M.R.**, Dimyati, M. & Gamal, A. (2025). Harmonizing street-view semantics and spatial predictors for dominant urban visual composition modelling. *Computational Urban Science*, 5(50). [\[DOI\]](#)
- [3] **Pradana, M.R.**, Wibowo, A. & Semedi, J.M. (2025). Multi-perspective evaluation of urban green views: Spatial and street-view data integration in Sudirman Central Business District, Indonesia. *Geomatics and Environmental Engineering*, 19(3), 91–114. [\[DOI\]](#)
- [4] **Pradana, M.R.**, Dimyati, M. & Semedi, J.M. (2025). Unveiling greenery and visual comfort: Integrating Green View Index and image segmentation in panoramic rural landscapes. *GeoScape*, 19(2), 143–159. [\[DOI\]](#)
- [5] **Pradana, M.R.** & Dimyati, M. (2024). Tracking urban sprawl: A systematic review and bibliometric analysis of spatio-temporal patterns using remote sensing and GIS. *European Journal of Geography*, 15(3), 190–203. [\[DOI\]](#)
- [6] Gamal, A., **Pradana, M.R.**, Wibisono, B.H., Navitas, P. & Aryal, J. (2025). Thematic fragmentation and convergence in urban flood simulation research: A 45-year bibliometric mapping. *Urban Science*, 9(7), 280. [\[DOI\]](#)

- [7] Semedi, J.M., **Pradana, M.R.**, Putera, D.A. & Rahatiningtyas, N.S. (2024). The impedance of East Jakarta road network to estimate fire station service areas. *Forum Geografi*, 38(2), 280–290. [DOI]

Conference Proceedings

- [1] Semedi, J.M., **Pradana, M.R.**, Permata, C.F., Pratama, R.A., Candra, M.A.N., Rahatiningtyas, N.S. & Dimiyati, M. (2026). Visualizing Sustainable Campuses: A Spatiotemporal Deep Learning Analysis of Eye-Level Greenery. *IOP Conference Series: EES*, 1648(1). [DOI]
- [2] **Pradana, M.R.** & Semedi, J.M. (2025). Effortless coastal monitoring: Unsupervised detection of shoreline alterations due to tin mining in Bangka Belitung. *IOP Conference Series: EES*, 1462(1). [DOI]
- [3] Ash-Shidiq, I.P., Supriatna, S., Darmawan, A., Warta, Z., Molidena, E., Valla, A., Hisan, Firdaus, M.I., Zakaria, N.A., Muafiroh, S., **Pradana, M.R.**, Putera, D.A. (2024). Biomass stock estimation using Landsat 8 imagery in Bukit Tigapuluh National Park, Riau. *IOP Conference Series: EES*, 1291(1). [DOI]

Under Review

- [1] **Pradana, M.R.** (*In review*). Your Academic Career Depends on Your Co-Authors: When Single-Author Papers Do Not Count in Indonesia's Academic Promotion System. *Journal of Academic Ethics*.
- [2] **Pradana, M.R.**, Gamal. A., & Aryal. J. (*In review*). Reframing Urban "Sustainable" Streetscapes: Evidence from Cross-City Transformation and Convergence. *City and Environment Interactions*.
- [3] **Pradana, M.R.**, Gamal. A., & Aryal. J. (*In review*). Interpreting 'Sustainable' Urban Streetscapes with Generative AI: Context-Rich vs. Generic Prompting. *Urban Informatics*.

RESEARCH EXPERIENCE

Research Assistant

May 2025 – Present

The Spatial Modeling and Analysis Research Group (SPARC), Dept. of Geography, Universitas Indonesia

- Co-led a university-funded grant (IDR 100 million) on climate resilience in urban parks, integrating street-view imagery and place attachment analysis.
- Mentored 3 undergraduate students across the full research cycle, from data collection and analysis to manuscript preparation.

Full-time Researcher & Assistant Editor of *Smart City Journal*

Oct. 2024 – Jun. 2026

SMART CITY Universitas Indonesia — Research & Community Engagement Unit

- Contributed to professional urban planning projects and co-led a university-funded research study on sustainable streetscapes (manuscript under review), alongside a multi-stakeholder urban flooding research project granted by INSPIRASI as GIS researcher.
- Managed the *Smart City Journal* editorial workflow, including manuscript handling and reviewer coordination.

Part-time Research Assistant

Jun. – Oct. 2024

SMART CITY Universitas Indonesia — Research & Community Engagement Unit

- Conducted GIS and remote sensing analyses using 3D urban data and digital building audits for a government-collaborated urban research project.
- Served as Project-in-Charge, coordinating project implementation and analytical deliverables for an urban research collaboration with government partners.

Student Researcher

Jun. 2023 – Oct. 2024

Department of Geography, Universitas Indonesia

- Conducted fieldwork and spatial analysis on urban fire management in Jakarta, resulting in 1 journal publication and 1 international conference presentation.

PROFESSIONAL EXPERIENCE

Cartographer — SEA, West Bangka Regency

Mar. – Dec. 2025

Center for Social and Environmental Management Research (PPSML), Universitas Indonesia

- Produced 50+ thematic and base maps at multiple cartographic scales for the Strategic Environmental Assessment (SEA) document.

Cartographer — Detailed Spatial Planning, Baradatu District

May – Aug. 2024

Center for Applied Geography Research (PPGT), Universitas Indonesia

- Compiled 20+ large-scale base and thematic maps at 1:5,000 for the district spatial planning (RDTR) document.

Cartographer — SEA, Bangka Belitung Islands Province

Mar. – Oct. 2024

Center for Social and Environmental Management Research (PPSML), Universitas Indonesia

- Produced 40+ thematic maps, including environmental sensitivity and infrastructure layers, for the SEA document.

Cartographer — Regional Spatial Planning, Hulu Sungai Tengah Regency

Jan. – Apr. 2023

Center for Applied Geography Research (PPGT), Universitas Indonesia

- Produced 40+ thematic maps at 1:50,000 scale for the regional spatial planning (RTRW) document in South Kalimantan.

PROJECTS

Research Projects

- Prompting Sustainability: A Comparative Analysis of How Generative AI Interprets and Visualizes Urban 'Sustainable' Streetscapes [[Project page](#)] Oct. 2025 – Present
Funded by Universitas Indonesia
- Cooling Cities, Warming Communities: Quantifying Climate Resilience in Urban Parks Through Anthropomorphic Mapping and Place Attachment [[Project page](#)] May 2025 – Present
Funded by Universitas Indonesia
- Spatial Modeling of Visual Element Proportion Diversity [Master's Thesis] [[Research deliverable](#)] Jun. 2024 – Jun. 2025
Conducted under SMART CITY UI and supported by INSPIRASI Grant Program

- Collaborative Mapping & Mitigation for a Resilient Capital City with Generative AI [\[Research deliverable\]](#) Jun. 2024 – Dec. 2025
Funded by PRIME Engineering and Universitas Indonesia Grant
- Level of Protection of Fire-Prone Areas in East Jakarta City [\[Research deliverable\]](#) Aug. 2023 – Jan. 2024
Funded by Faculty of Mathematics and Natural Sciences, Universitas Indonesia

Government Projects

- Innovative Government Awards 2025 — Ministry of Home Affairs (*Lead data scientist*) Sep. 2025 – Nov. 2025
- Strategic Environmental Assessment (SEA) — West Bangka Regency Mar. 2025 – Dec. 2025
- Innovative Government Awards 2024 — Ministry of Home Affairs (*Lead data scientist*) Sep. 2024 – Oct. 2024
- Strategic Environmental Assessment (SEA) — Bangka Belitung Islands Province Mar. 2024 – Oct. 2024

OPEN-SOURCE TOOLS & DATASETS

Project Page

- Prompting Sustainability** 2025 - 2026 
- Cooling Cities, Warming Communities** 2025 - 2026 

Open Repositories

- M2F Playground** — All in one easy to access Notebook repo for Mask2Former inference 2026 
- GSV Panorama Downloader** — Notebook repo for batch downloading Google Street View panoramas 2025 
- GSV Downloader by Links** — Notebook repo for downloading Street View imagery via direct links 2025 
- Facade Dominant Color Extractor** — Notebook repo for extracting color information of building facade from street-view imagery 2025 

Open-Access Datasets

- genai_sustainablestreetscape (v2)** — Cross-city corpus of 4,186 paired images (raw + GPT-5-generated counterparts) from Jakarta (567), Melbourne (832), and Singapore (694). 2026 
- Sustainable Streetscape Prompt-Specificity Dataset (v1)** — 100 Jakarta GSV scenes × 3 treatments (raw, context-rich, non-context via GPT-4o), each paired with semantic segmentation masks. 2026 

TEACHING EXPERIENCE

- Guest Teaching — Data Management and Analysis** Apr. – May 2026
Master Programme of Urban Planning, Universitas Indonesia
 - Delivered lectures on database development, statistical analysis, and spatial analysis for graduate urban planning students.

Teaching Assistant — Introduction of GIS & Remote Sensing

Jun. – Aug. 2024

Department of Geography, Universitas Indonesia

- Led practical sessions for 30+ undergraduates across six GIS and remote sensing modules.

PEER REVIEW SERVICE

GeoJournal, Springer Nature	2026
Journal of Geovisualization and Spatial Analysis, Springer Nature	2026
Discover Environment, Springer Nature	2026
Journal of Mountain Science, Springer Nature	2026
Scientific Reports, Nature Publishing Group	2026
Frontiers in Marine Sciences, Frontiers	2026
Urban Forestry & Urban Greening, Elsevier	2025
Transactions in GIS, Wiley	2025

CONFERENCE PRESENTATIONS

- **The 2nd International Conference on Anthropocene, Global Environmental Change & Powerful Geography** (Surakarta, Indonesia) Aug. 2024
Effortless Shoreline Detection in Tin-Mining Areas. [Best Presenter Award]
- **The 1st Aceh International Conference in Geography & Environmental Education** (Aceh, Indonesia) Nov. 2023
The Impedance of East Jakarta Road Network to Estimate Fire Station Service Areas.
- **The 2nd International Conference of Science & Applied Geography** (Depok, Indonesia) Nov. 2022
Biomass Stock Estimation Using Landsat 8 Imagery in Bukit Tigapuluh National Park.

HONORS & AWARDS

- 2025 **Ranked 1st in Cohort, M.Sc. Geography** — Summa Cum Laude (Faculty Level)
- 2024 **Ranked 1st in Cohort, B.Sc. Geography** — Cum Laude (Department Level)
- 2024 **Fast-Track Program Candidacy, B.Sc. - M.Sc. Geography** — selected based on academic excellence during undergraduate study
- 2024 **Best Presenter Award, ICoAGPG 2024**
- 2022 **2nd Place, National Economics Paper Competition (IMIE FEB UKSW)**

TECHNICAL SKILLS

- **Urban Analytics:** Street-view imagery, object extraction through computer vision, urban data science, geospatial data science, spatial analysis
- **Programming:** Python (familiar), R (familiar)
- **Geospatial Platforms:** ArcGIS Pro, QGIS, Google Earth Engine
- **Research Computing:** Google Colab, Jupyter Notebook, VS Code, Git, GitHub, LaTeX

- **Quantitative Methods:** Statistical analysis, machine learning, computer vision, dimensionality reduction, systematic literature review, bibliometric analysis